

Update on 2026 tracking efficiencies

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Tag-and-probe method

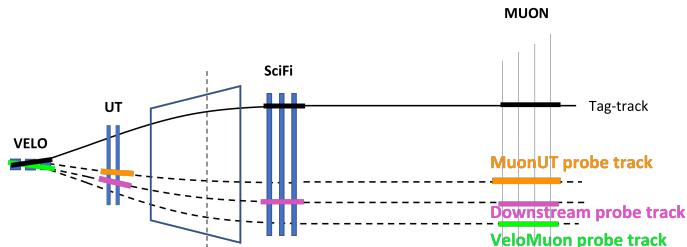


Figure by Rowina

- Tag-and-probe method with $J/\psi \rightarrow \mu^+ \mu^-$
- Fully reconstruct one muon...
- ... and partially reconstruct the other muon
- Match hits in specific sub-detector with partially reconstructed track

$$\epsilon_{\text{track}} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{failed}}}$$

Strategy for tracking efficiency determination:

- Combine efficiencies of two methods:
 - VeloMuon: SciFi efficiency
 - Downstream: Velo efficiency
 - Combined: $\text{Velo} \times \text{SciFi}$ efficiencies
- Cross check with long track efficiency from the MuonUT method
 - Main systematic

Information for analysts:

- v0 tables available on GitLab [here](#)
- Muon tracking efficiency corrections and hadronic corrections
- Currently preliminary tables are available for 2024 and 2025
- Note: Hadronic corrections just updated, plan to present next week

Tracking efficiencies are determined using TrackCalib2

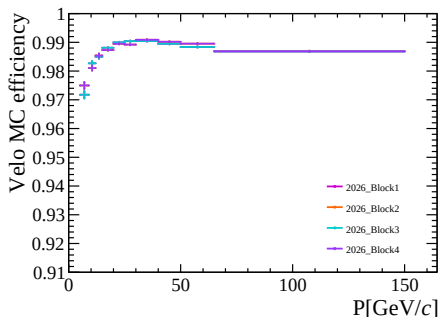
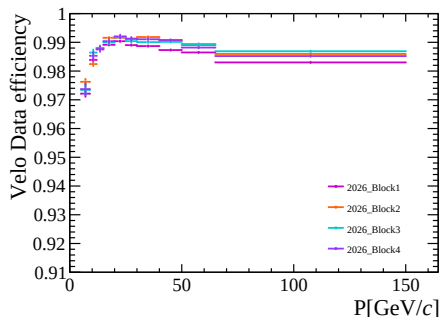
- GitLab repository [here](#)
- Mass fit done on tuples from AP to extract efficiencies
- Note: Analysts should not have to run TrackCalib2 themselves, unless the released numbers are not suitable (need different binning, etc)

Today I will present 2026 tracking efficiencies:

- Note: Expected 2026 MC is used
- Split into blocks by sprucing campaign
 - Block 1: Sprucing26c1 MagDown
 - Block 2: Sprucing26c1 MagUp
 - Block 3: Sprucing26c1 MagDown
 - Block 4: Sprucing26c1 MagUp

2026 tracking efficiencies

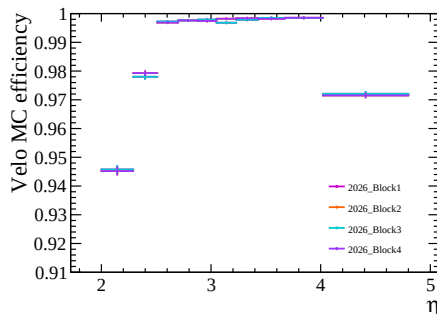
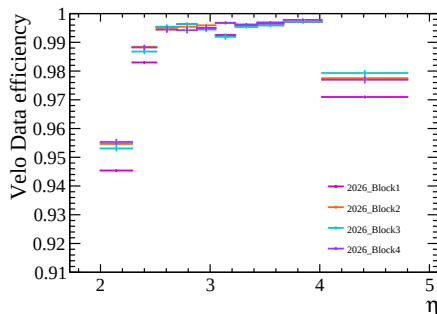
Comparison of tracking efficiencies in 2026 data blocks



Velo efficiencies agree reasonably well between data and MC in p ...

2026 tracking efficiencies

Comparison of tracking efficiencies in 2026 data blocks



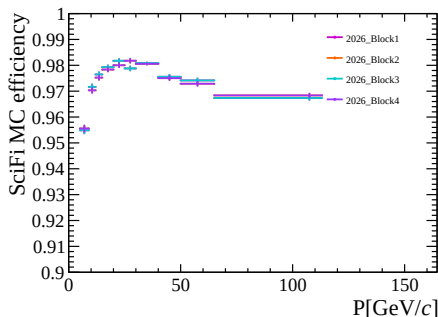
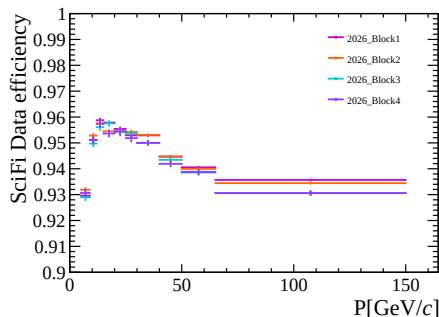
... but in η some features are seen at low and high η after 2026 Block 1

Velo changes during data taking

- It's unclear why the Velo efficiencies are higher in Blocks 2, 3, 4
- Magnet flip happened on 25th March 2026
- From Run meeting Velo minutes on 20th March 2026:
 - *"Tuesday. HV recipes updated. STANDBY2 with 400V for inner sensors and 200V for outer sensors. M30 VP2 with 600V at ready"*
- From Run meeting Velo minutes on 24th March 2026:
 - *"HV recipes updated with M30/M31 VP2 and VP0 at 600V"*
 - *"... After some investigation, M30 VP2-0 was confirmed to be damaged. HV recipes rollback to 400V inner sensors and 200V outer sensors."*
- Furthermore, several ASICs failed during this period...
- ... but change in HV could explain the higher efficiencies

2026 tracking efficiencies

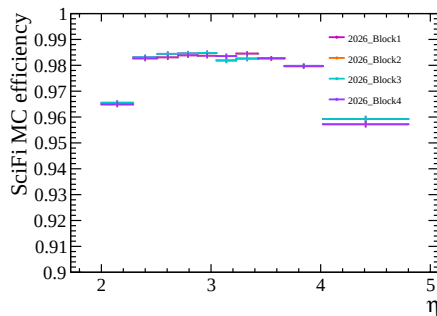
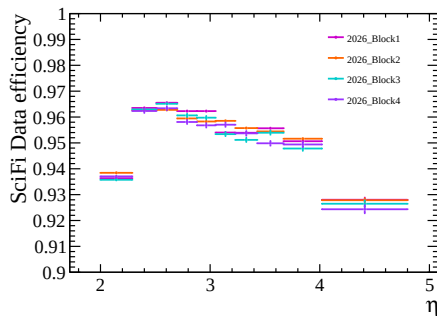
Comparison of tracking efficiencies in 2026 data blocks



SciFi efficiencies looks quite different between data and MC...

2026 tracking efficiencies

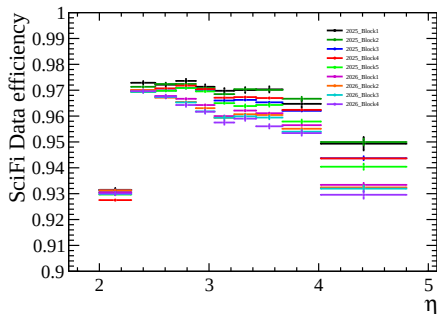
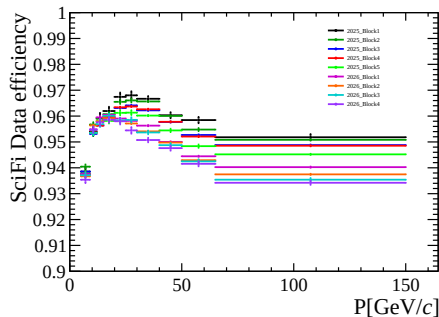
Comparison of tracking efficiencies in 2026 data blocks



SciFi efficiencies look quite different between data and MC...

SciFi radiation damage?

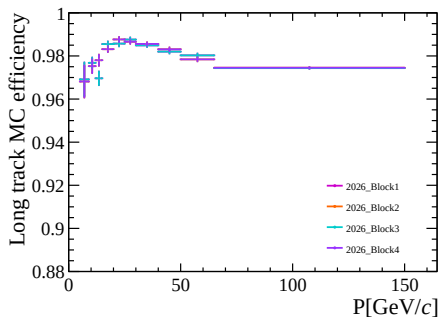
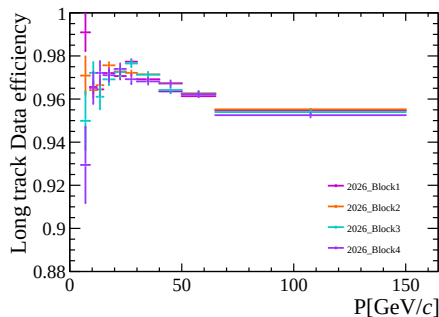
Plot from RTA WP4/5 meeting on 4th June 2026:



- Clear signs of radiation damage at higher η ...
- ... but it is unclear what caused the drop between 2025 and 2026

2026 tracking efficiencies

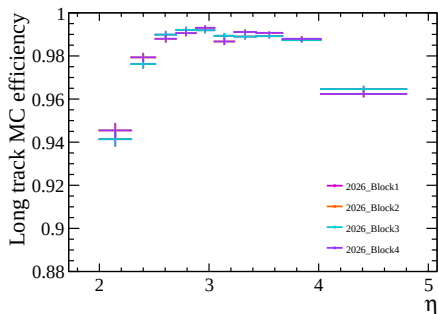
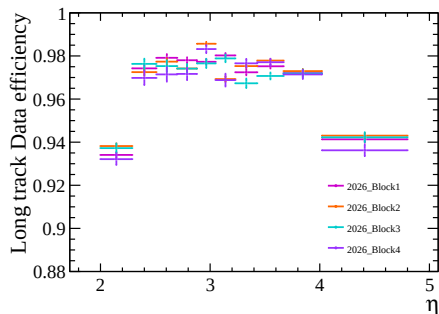
Comparison of tracking efficiencies in 2026 data blocks



MuonUT tracking efficiencies have consistent trend between data and MC

2026 tracking efficiencies

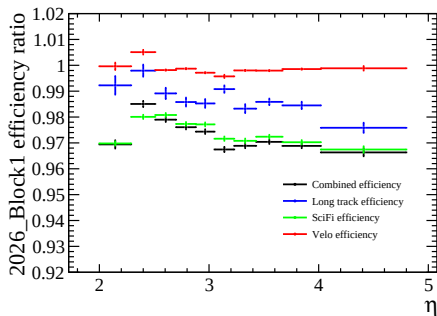
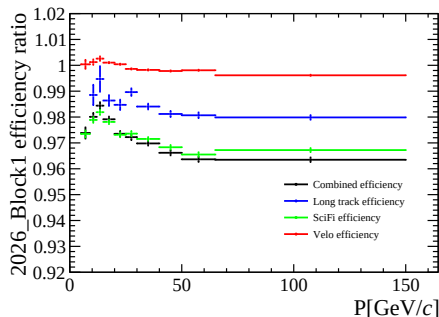
Comparison of tracking efficiencies in 2026 data blocks



MuonUT efficiencies have consistent trend between data and MC

2026 tracking efficiency ratios

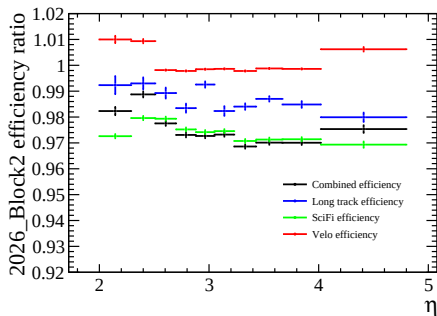
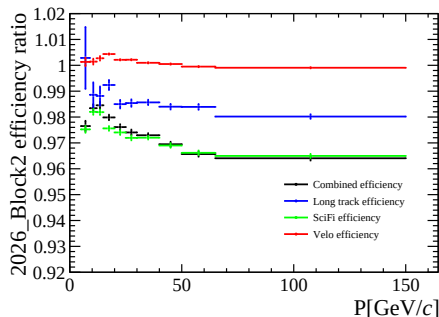
Comparison of tracking efficiency ratios in 2026 Block 1



No surprises in 2026 Block 1

2026 tracking efficiency ratios

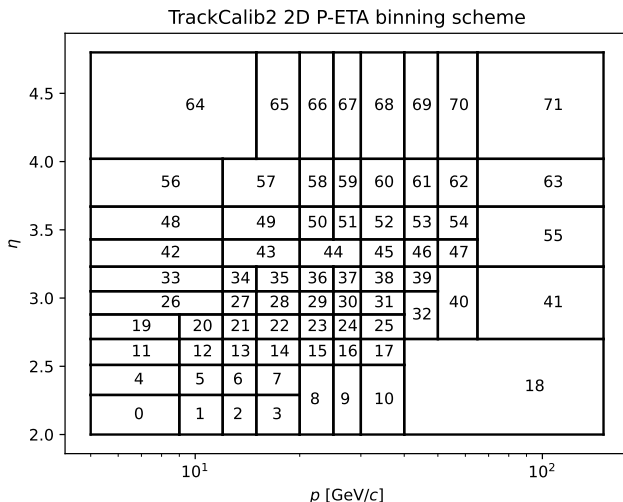
Comparison of tracking efficiency ratios in 2026 Block 2

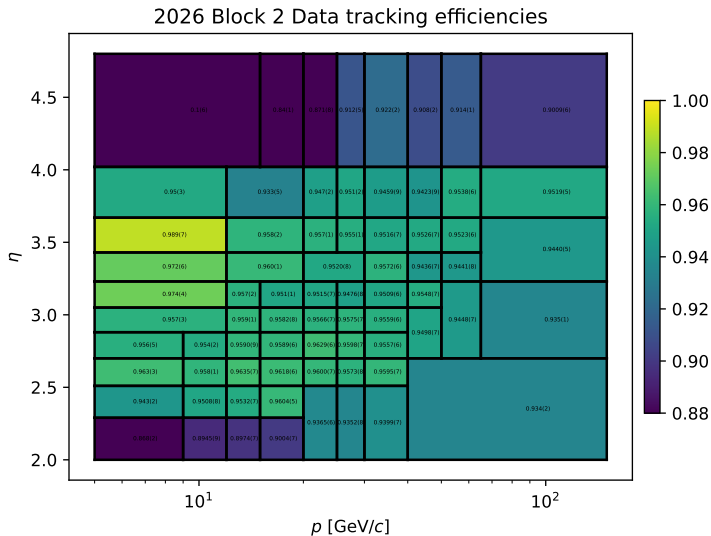


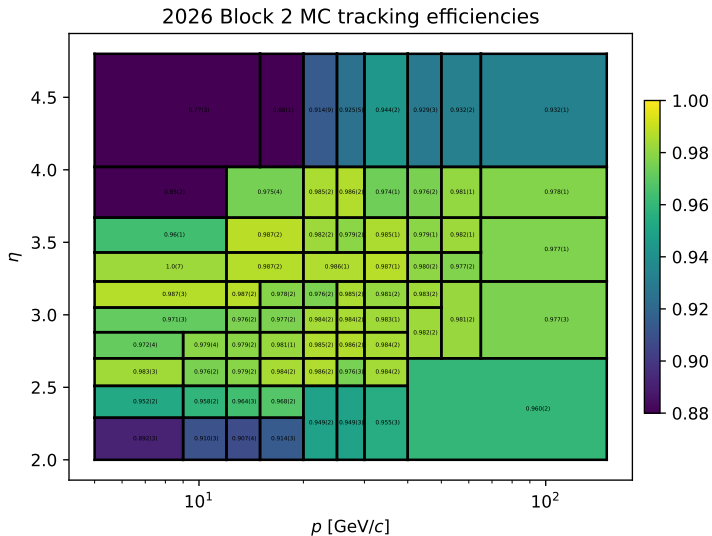
2026 Block 2 shows some discrepancies in Velo efficiencies

2D binning

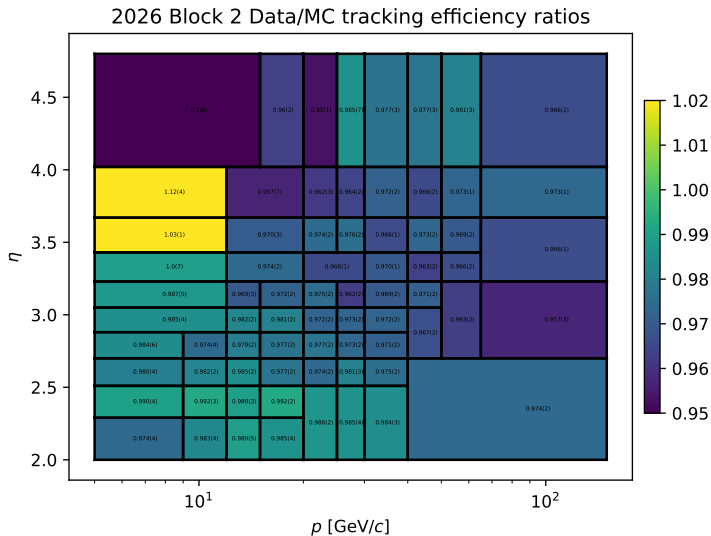
2D binning in p vs η with 72 bins:





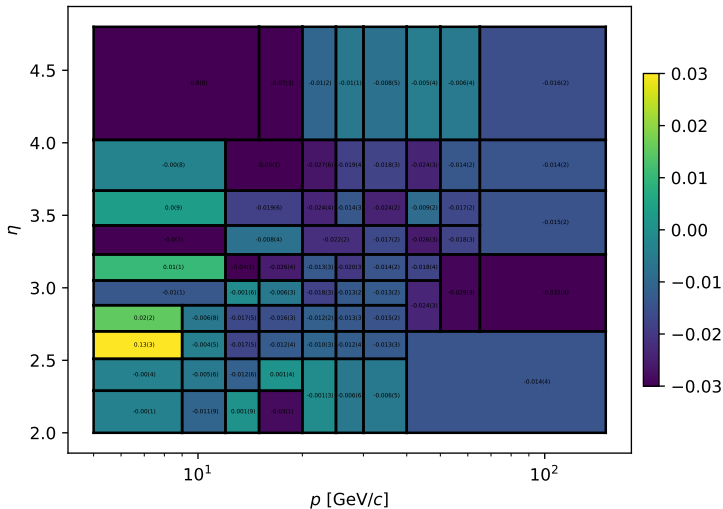


2D binning



2D binning

2026 Block 2 Data/MC Combined vs MuonUT ratio difference



We have prepared 2026 tracking efficiency tables, but:

- Expected 2026 MC is used
- Large discrepancies in SciFi seen, possibly due to radiation damage
- Issues with Velo efficiencies at high and low η to be understood
- Feedback/comments welcome!

Currently working on a new release of 2024 efficiencies:

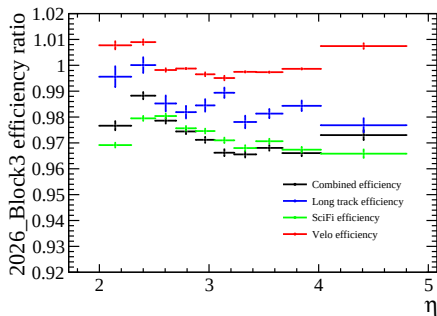
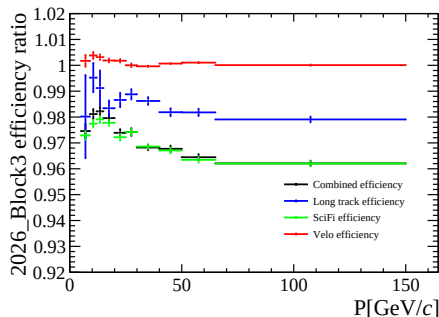
- ① Larger MC sample
- ② Separate corrections for $\mu^\pm$, μ^+ , μ^-
- ③ New 2D binning

Thanks for your attention!

Backup

2026 tracking efficiency ratios

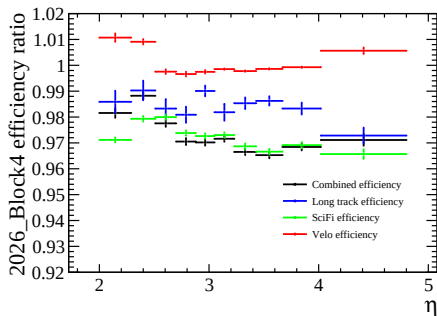
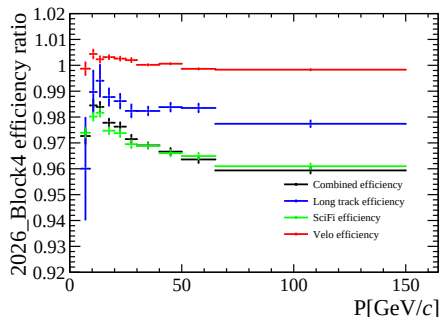
Comparison of tracking efficiency ratios in 2026 Block 3



2026 Block 3 shows some discrepancies in Velo efficiencies

2026 tracking efficiency ratios

Comparison of tracking efficiency ratios in 2026 Block 4



2026 Block 4 shows some discrepancies in Velo efficiencies